

Assignment 4 - Mission to Mars: Iteration and Loops

Submission Date: Tuesday 6th October 2026 11.59pm on Canvas

This assignment practices iteration introduced in the Week 4 notebook, 4_01_Iteration.ipynb. Your rover has landed on Mars. Mission Control needs you to program its exploration route, track samples, avoid hazardous terrain, etc.. Complete five short programming tasks to prepare the rover for its first mission, using short, simple loops and building on the data structures from Assignment 3. Work one part at a time and use the notebook examples for help. Use the requested loops rather than manually repeating print statements or typing calculated answers. Nested loops, user-defined functions, additional packages, objects, or other complications, are not required.

Complete the work in a new Google Colab notebook. Put your name, course number, and assignment title at the top, label every part clearly, run every code cell, and leave all output visible. Use descriptive snake_case names and include units where appropriate.

Table 1 outlines the required assignment components.

Component	Requirements	Points
1. Explore the landing site	Create exploration_route = ["Base Camp", "Crater Rim", "Rock Field", "Ice Deposit"]. Use a for loop to print each stop on a separate line with a short message, such as "Visiting Base Camp". Then use a second loop with enumerate(exploration_route) to print each stop and its index. In one sentence, explain why the first index is 0.	20
2. Number sample containers	Use a for loop and range() to print container numbers 1 through 6 inclusive. Then use another for loop with range(start, stop, step) to print 2, 4 and 6. Write your predicted output for each loop in a text cell before running it. In one sentence, explain why the stop value is excluded.	20
3. Inspect the rover	Create rover_status = {"name": "Pathfinder", "battery_percent": 100, "operational": True}. Use a for loop with .items() to print each key and its value together. Then use a separate loop with .keys() to print only the keys, and another with .values() to print only the values. Label the three sets of output so they are easy to compare.	20
4. Count collected samples	Create sample_counts = [12, 8, 15, 5], representing rock samples collected during four collection periods. Start total_samples at 0. Use a for loop to add each count to total_samples and print the running total inside the loop. After the loop, print the final total with a descriptive label. In one sentence, explain why total_samples must be initialized before the loop. Use your loop to calculate the total rather than sum().	20
5. Perform launch checks	Start check_number at 1. Use a while loop to print "Launch check" and the current number for checks 1 through 5 inclusive. Increase check_number by 1 inside the loop. After it ends, print "Rover ready!". In one sentence, explain what would happen if you removed the counter update. Do not run a version that omits the update.	20

Table 1 Assignment components and requirements.

Submit the completed .ipynb file online. Before submitting, restart the runtime and run the notebook from top to bottom. Check that all five parts, requested outputs, predictions and short explanations are present, and no unexpected errors remain. Each part should define its own starting data and counters so it runs reliably. Check your indentation and confirm that the while loop finishes. All locations are fictional. Where a prediction is requested, briefly state whether your actual output matched it and explain any difference.

Also submit a .PDF file of your notebook (or a .PDF of screenshots), for easy review.

Table 2 outlines the assessment criteria. Component points total 100; the criteria describe how the work is assessed across those components.

Criterion	Evidence	Weight
Correctness / completeness	All five parts are present; loop outputs, container numbers, running totals, launch checks and the final visited count are correct.	40%
Loop selection / use	The requested for and while loops, range(), enumerate(), dictionary methods, continue and break are used appropriately. Counters are initialized and updated correctly.	25%
Documentation / readability	Names preferably follow snake_case, indentation and output labels are clear, and the requested predictions and short explanations are understandable.	15%
Testing / output	All requested loops are run and their outputs are visible. Predictions are compared with actual output; the while loop terminates and loop-control behavior is demonstrated.	15%
Organization / submission	The notebook is clearly labeled, logically ordered, complete, and submitted in the required formats.	5%

Table 2 Assessment criteria.

On this particular assignment, you are not allowed to use AI, otherwise it defeats the point of trying to learn this content. All the operations you need are covered in the Week 4 iteration notebook, and you are allowed to read the Python documentation for extra background. The knowledge from this assignment is cumulative and necessary to succeed in later weeks of the course (and especially in the pencil and paper exam where you cannot use AI). Undisclosed AI-generated work, fabricated material, or code that the student cannot explain may constitute an Honor Code violation.